

*İlhan Günel
Figen Batioğlu*Eye Clinic, Faculty of Medicine,
University of Ankara, Turkey

Effect of Pre-Enucleation Irradiation on the Survival of Patients with Uveal Melanoma

Key Words

Enucleation
Irradiation
Pre-enucleation irradiation
Uveal melanoma

Abstract

The effect of pre-enucleation irradiation on the survival of patients with uveal melanoma was investigated in 140 patients. Of the patients, 42 received pre-enucleation cobalt external radiotherapy 2,000 cGy, in five fractions between 1981–1991. The control group consisted of 98 patients with uveal melanoma treated by enucleation alone between 1964 and 1994. The great majority of tumors in each group were confined to the choroid. The mean follow-up period was 58.3 months in the irradiated group and 61.6 months in the enucleation group. A comparison of the Kaplan-Meier survival curves of the patients managed by pre-enucleation irradiation and enucleation indicated no significant difference in survival between the groups. After 5 years of follow-up, the survival rates for the preoperatively irradiated group and the control group were, 86.5 and 86.8%, respectively. In both groups, women had a significantly better survival than men. It is more likely that pre-enucleation irradiation does not appear to improve survival in patients with uveal melanoma. The reason for this may be the micrometastases that occur before treatment.

Introduction

Although it has been generally accepted that enucleation is the only acceptable treatment for posterior uveal melanomas, it has become controversial in the past 2 decades [1]. In 1942, Lehrfeld [2] showed that early enucleation had no effect on metastasis and tumor-related mortality. In 1978, Zimmerman et al. [3, 4] reported similar results. They reported that the tumor death rate following enucleation increased within 2 years [3, 4]. Surgical trauma during enucleation may enhance hematogenous dissemination of tumor cells, the immune response of the host may change after excision of the tumor, and the tumor cells which were disseminated before enucleation may increase after surgery. Jensen [5] found tumor death rate following enucleation of

about 50%, and half the patients succumbed within the first 3 years. Recent studies have led to alternative enucleation modalities, and a 'no touch' technique has been developed to avoid dissemination of the cells during enucleation [6] but has not received wide acceptance. The value of pre-enucleation irradiation in the management of uveal melanoma is uncertain. Studies in mice show that the probability of developing metastasis after pre-enucleation radiotherapy is low [7]. Although pre-enucleation radiotherapy seems to decrease the degree of mitotic activity in melanoma cells and to inhibit in vitro growth of tumor cells [8], available data suggest that it does not improve survival.

In this study, the effect of pre-enucleation irradiation on the survival of patients was compared with a control group of patients treated by enucleation alone.

KARGERE-Mail karger@karger.ch
Fax +41 61 306 12 34
www.karger.com1998 S. Karger AG, Basel
0030-3755/98/2124-0231 \$ 15.00/0Prof. İlhan Günel
Cinnah Cad
Nihat Erim İşhane 37/20
Çankaya, Ankara (Turkey)
Tel. 90 312 4403837

Materials and Methods

All patients were examined at the Tumor Department of the Eye Clinic, Ankara University. The diagnosis of uveal melanoma was established based on indirect ophthalmoscopy, fundus photography, fluorescein angiography and ultrasonography. All patients had a complete physical examination before surgery.

Between 1964 and 1994, 240 patients with uveal melanomas were treated by enucleation. From 1981 to 1991, 62 patients received pre-enucleation irradiation. During this period, patients were treated at random either by enucleation or by pre-enucleation irradiation. This study included patients who could be followed-up at least 6 months in both groups.

The first group consisted of 42 patients with uveal melanomas who received pre-enucleation irradiation between 1981 and 1991. Of the 42 patients, 24 were males and 18 females. Their mean age was 44.8 years (range 18–71 years). Six (14%) were ciliary body tumors and 36 (86%) were choroidal melanomas. Thirty-two of the patients had spindle, 3 had mixed and 7 had epithelioid cell tumors. The tumor diameter was 10 mm or less (T_1) in 28, 11–15 mm (T_2) in 13, and 16 mm or greater (T_3) in 1 patient. Mean follow-up was 58.3 months (range 6–151 months).

The control group included 98 patients whose eyes were enucleated because of an uveal melanoma. Of these patients, 60 were males and 38 were females. Their mean age was 45.9 years (range 10–79 years). Fourteen (14%) of the 98 melanomas were ciliary body and 84 (86%) were choroidal melanomas. Eighty-two of the tumors were spindle cell melanomas whereas 12 were mixed and 4 were epithelioid lesions. The tumor diameter was T_1 in 51, T_2 in 32 and T_3 in 15 patients. The patients could be followed 6–207 months (mean 61.6 months).

Pre-enucleation irradiation was performed using cobalt-60 external radiotherapy on the 5 × 5 mm orbit area anteriorly. The patients were given a total dose of 2,000 cGy, as 400-cGy daily fractions for 5 days. Enucleation was performed using a standard enucleation technique. In the patients who received pre-enucleation irradiation, enucleation was performed usually within 1–5 days of completion of radiation.

Patients were followed up at 6-month to 1-year intervals postoperatively. Follow-up examination consisted of ophthalmoscopy of the remaining eye, inspection of the socket, and investigation for systemic metastases. Patients were followed up until death or December 1994. The survival status of the patients failing their appointment was determined by contacting their family members. The date and cause of death was verified from medical records or death certificates generated during the terminal illness. Uveal-melanoma-related death was taken into consideration. In the irradiation group, there were 4 deaths (9%) from metastases, and in the enucleation group metastasis-related deaths occurred in 12 (12%).

The χ^2 test was used to compare the irradiated group with the control group for nonparametric variables such as gender, tumor location, tumor size and cell type, and Student's *t* test for age which was a parametric variable. Statistical analysis to study the effect of preoperative irradiation on survival was performed using Kaplan-Meier survival curves and Cox proportional hazard analysis.

Table 1. Patient and tumor characteristics between groups

	Enucleation n = 98	Pre-enucleation irradiation, n = 42	p value
Average age, years	45.9±13.4	44.8±13.5	0.649
Gender			0.651
Men	60	24	
Women	38	18	
Location			0.792
Ciliary body	14	6	
Posterior uvea	84	36	
Tumor size			0.0659
T_1	51	28	
T_2	32	13	
T_3	15	1	
Cell type			0.0327
Spindle	82	32	
Mixed	12	3	
Epithelioid	4	7	

Results

The data on gender, mean age, tumor size classification, cell type and tumor location for the irradiated group and the enucleation group are shown in table 1. There was no statistically significant difference between the two groups in gender, mean age, tumor location and tumor size ($p > 0.05$), but, in the cell type, the two groups differed significantly.

Metastatic disease occurred in 4 of 42 patients (9%) treated with pre-enucleation irradiation and 12 of 98 enucleation patients (12%). The mean onset of metastases was 30.8 months in the irradiation group, while it was 54.1 months in the enucleation group. The onset of metastatic disease seems to occur sooner in the irradiation group, but it was not statistically significant ($p = 0.1803$). The mean age at the onset of metastases was 50.8 years in the enucleation group while it was 35.5 years in the irradiation group. Both in the irradiation and enucleation groups metastatic disease occurred mostly in men. The rates were 75 and 91%, respectively, and the two groups did not differ significantly ($p = 0.45$).

The 5-year survival rates for the preoperatively irradiated group and the enucleation group were 86.5 and 86.8%, respectively. The mean survival time was 135.5 months (95% confidence interval, 121.1–149.9) in the pre-enucleation irradiation group while it was 169.7 months (95% confidence interval, 150.7–188.7) in the enucleation group. No significant difference was found with the log-rank test ($p = 0.25$). According to Kaplan-Meier, the survival esti-

Fig. 1. Kaplan-Meier survival curves of the pre-enucleation and the enucleation groups ($p = 0.25$).

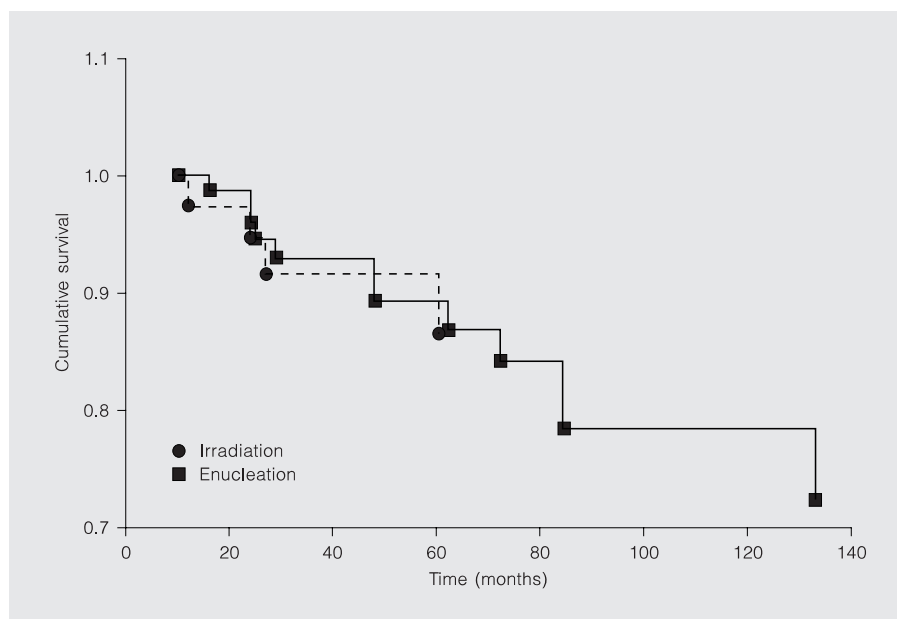
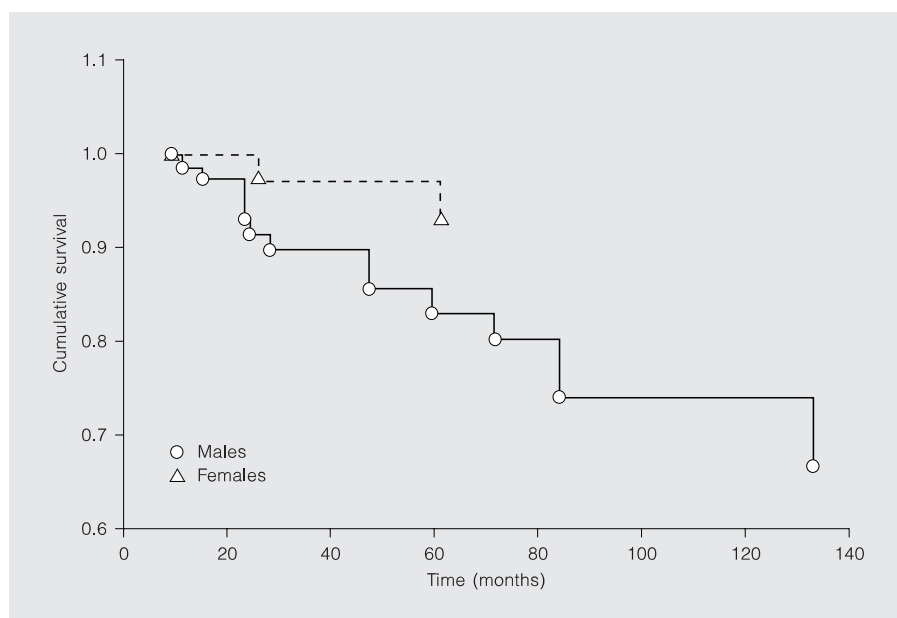


Fig. 2. Kaplan-Meier survival curves for women and men in the total group (140 patients; $p = 0.03$).



mate curve of the preoperatively irradiated group and the enucleation group is shown in figure 1. Both in the irradiation and in the enucleation group ($n = 140$) women had a significantly better survival than men ($p = 0.03$; fig. 2). The 5-year survival rates for women and men were 92.8 and 82.9%, respectively. This rate decreased to 66.8% at 133 months of follow-up in men while it did not significantly change in women.

The Cox proportional hazard analysis was used to study the effect of pre-enucleation irradiation on metastases ad-

justing for prognostic covarities, such as tumor size and cell type. No association was found with the tumors of increased diameter ($p = 0.89$), and malignant cell type ($p = 0.58$). This might be due to the greater number of T₁ and spindle-cell-type tumors in our study.

Discussion

Uveal melanoma is the most common primary intraocular malignancy, and it is the main primary intraocular disease that can be fatal in adults. It is most often diagnosed in the 6th decade of life and is somewhat more common in males. In our study, average ages in enucleation and pre-enucleation irradiation groups were 45.9 and 44.8 years. In both groups, there was male predominance. It seems that uveal melanoma tends to be diagnosed nearly 1 decade earlier in our country. The risk of developing the disease at an earlier age in the Mediterranean region may account for this.

A variety of methods for treating uveal melanoma have been developed with the aim of conserving a useful eye, but these can be justified only if and when they do not increase the risk of metastatic disease compared with enucleation. It has been suggested that dissemination of tumor emboli at the time of enucleation may be responsible for the peak incidence of metastatic deaths 2 years after surgery [3, 4, 9]. Manschot et al. [10] concluded that any metastatic death occurring less than 7 years after treatment is the result of tumor dissemination before treatment. Altering the enucleation procedure or delivering pre-enucleation irradiation may reduce the incidence of dissemination of these malignant cells. Reductions of 90 and 95% in the proliferation activity of melanoma cells were demonstrated after 8- and 29-Gy irradiation doses, respectively, by an *in vitro* study [11]. Fournier et al. [12] delivered 50 Gy to hamsters with Greene melanomas injected into their chambers, and there metastasis and mortality decreased compared with controls. For human uveal melanomas, a significant reduction in the proliferative activity/mitotic rate was found after preoperative irradiation with 8 [13] and 20 Gy [14].

In this study, we observed that pre-enucleation radiotherapy aimed at reducing tumor seeding did not seem to have reduced the risk of metastatic disease. The 5-year survival rates for the preenucleation radiation and the enucleation group were 86.5 and 86.8%, respectively. Augsburger et al. [15] found a cumulative 5-year survival probability of 57.9% for 29 patients in the enucleation group versus 63.9% for 29 patients in the pre-enucleation radiotherapy group. They also failed to identify any beneficial effect of pre-enucleation radiotherapy on survival during the first 5 years after enucleation. Char et al. [16] studied a small series of nonrandomly selected patients and found that preoperative irradiation with 20 Gy had a worse prognosis compared with patients treated by enucleation alone. Bornfeld et al. [17] reported survival data on an uncontrolled prospective group of 80 patients with primary choroidal and ciliary

body melanoma treated with pre-enucleation radiotherapy, and no beneficial effect could be found in terms of better survival. In a similar study, Kreissig et al. [18] studied the survival experience of an uncontrolled retrospective group of 26 patients with choroidal melanoma treated with pre-enucleation radiotherapy. The mortality rate was 34.6% after an average follow-up of 4.5 years, and they concluded that pre-enucleation radiotherapy had no beneficial effect on life expectancy. More recently, Luyten et al. [19] also found no long-term difference in the survival of uveal melanoma patients treated by two fractions of 4 Gy before enucleation compared with patients treated by enucleation alone.

Approximately 50% of patients diagnosed with uveal melanoma will develop metastasis within 10–15 years after enucleation. The most long-term studies of survival following enucleation derive from Denmark and Finland. In Rii-vio's [20] study the 5-, 10- and 15-year survival rates based on melanoma-related deaths were 65, 52 and 46%, respectively. Jensen [5] evaluated survival at least 25 years after enucleation in Danish patients with uveal melanoma. The majority of patients had died (82%); 51% of those deaths were due to metastasis. In our study, the survival rate in the enucleation group was 86.8% after 5 years. After 15 years, the survival rate decreased to 72.4%. In the irradiation group, the survival rate was 86.5% after 5 years and remained the same, but the number of patients was too small, and the follow-up period was too short to give reliable rates. The reason why pre-enucleation radiation does not decrease uveal melanoma mortality may be due to micrometastatic disease which is present before irradiation and which is not affected by local irradiation of the primary tumor. Rousseau et al. [21] reported that histologic examination of the pathologic sections showed no effects of the preoperative radiotherapy treatment.

Although no significant difference was found between women and men [5, 22, 23] in several studies, we found a significant difference in survival between men and women. After 5 years, the survival rate was 92.8% in women and 82.9% in men. A better survival for women was also confirmed by Folberg et al. [24] and Luyten et al. [19].

In conclusion, the disease tends to occur at an earlier age in our country. No long-term beneficial effect on survival could be found after pre-enucleation cobalt external irradiation, and women had a better survival than men.

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